

Powers and Exponents

Flagship

Lesson 6-1-flagship

Name: _____

Date: _____

Class: _____

SOUND STUDIO MISSION

| The Doubling Mixboard

You are the lead sound engineer at Amplitude Studios. Every time you flip a switch on the new mixboard, the volume doubles — that is the power of exponents. The headline act goes on in one hour, and you must master powers before the speakers can be safely dialed in.

Key Vocabulary

Level 1 support

Picture first, then the word, then a plain-language meaning. Say each word out loud.

$$5^3$$

$$5 \times 5 \times 5$$

In 2^3 , the small 3 means multiply 2 by itself 3 times: $2 \times 2 \times 2 = 8$

Exponent

A small number that tells how many times to multiply the number by itself.



base
bottom side

In 5^2 , the base is 5 — it is the number being multiplied: $5 \times 5 = 25$

Base

The number that gets multiplied by itself.

$$5^3$$

$$5 \times 5 \times 5$$

$10^3 = 10 \times 10 \times 10 = 1,000$ — read as '10 to the third power' or '10 cubed'

Power

A number written with a base and an exponent, like 2^3 .

$$3x + 5$$

no equal sign

Evaluate 3^4 : write $3 \times 3 \times 3 \times 3$, then multiply step by step: $9 \times 9 = 81$

Evaluate

To find the value of an expression.

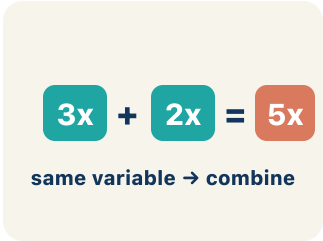

$$4x$$

coefficient = 4

In $3x$, the 3 is the coefficient — if $x = 4$, then $3x = 3 \times 4 = 12$

Coefficient

The number in front of a letter, like the 3 in $3x$.


$$3x + 2x = 5x$$

same variable $\rightarrow$ combine

$4x$ and $2x$ are like terms (both have x); $4x$ and $4x^2$ are NOT like terms (different powers)

Like terms

Terms with the same letter, like $2x$ and $5x$.

Key Ideas & Notes



WHAT WE'RE LEARNING TODAY

I can write and evaluate numbers in exponent form using a base and a power.

1 Notice

You're a sound engineer at a music studio.

2 Key idea

I can write and evaluate numbers in exponent form using a base and a power.

3 Words I'll use

Exponent

Base

Power

Evaluate

Coefficient

Like terms



Think About It

- What happens to the volume each time you flip a switch?
- Why is 2^3 equal to 8, not 6?
- How is repeated multiplication different from repeated addition?

See the notes in action: watch one worked all the way through, then try the next with the same steps.



I do — watch

Follow each step as your teacher solves it.

Problem: What is the value of 4^3 ?

- A. 64
- B. 12
- C. 43
- D. 7

Step 1 $4^3 = 4 \times 4 \times 4 = 64.$



Answer: A. 64

 We do — together

Solve this one with your class using the same steps.

Problem: What is the value of 6^2 ?

A. 36

B. 12

C. 62

D. 8

Step 1

Step 2

Answer:

 You do — your turn

Now try one on your own. Show every step.

Problem: Which expression shows 5^3 as repeated multiplication?

A. $5 \times 5 \times 5$

B. 5×3

C. $5 + 5 + 5$

D. $3 \times 3 \times 3 \times 3 \times 3$

Show your work:

Turn & Talk — Discussion Points

Level 1 support

Level 2

Talk with a partner about each prompt. If you need help getting started, use the Level 1 sentence stems. Ready for more? Try the Level 2 question.

LAUNCH

Each switch flip doubles the volume: 1, then 2, then 4, then 8. How is each number related to the one before, and how could you write 2 times 2 times 2 a shorter way?

Level 1 support

Start here: Each step multiplies by 2 again, so repeated multiplication can be written as a power like 2 cubed.

Need a hint?

Sentence starters (optional)

Each number is the one before it times ____ , so it is repeated ____ .

Cada número es el anterior por ____ , así que es ____ repetida.

2 times 2 times 2 can be written as 2 to the ____ power.

2 por 2 por 2 se puede escribir como 2 a la ____ potencia.

Word bank: base exponent power evaluate double

Listen for: Students see each volume is the previous one times 2 and write $2 \times 2 \times 2$ as 2 cubed, naming 2 as the base and 3 as the exponent.

Level 2

Predict the volume after 6 switch flips and write it as a power. Explain how the exponent connects to the number of flips.

- After 6 flips the volume is ____ , which I write as ____ .
- The exponent matches the number of flips because ____ .

EXPLORE

In the power 2 cubed, which number is the base and which is the exponent? What job does each one do when you evaluate it?

Level 1 support

Start here: The base is the factor being multiplied; the exponent tells how many times to use it.



Need a hint?

Sentence starters (optional)

The base is ____ and it is the number being ____.

La base es ____ y es el número que se está ____.

The exponent is ____ and it tells how many ____.

El exponente es ____ e indica cuántas ____.

Word bank: **base** **exponent** **power** **evaluate** **factor**

Listen for: Students identify the base as the repeated factor and the exponent as the count of factors, and evaluate 2 cubed as $2 \times 2 \times 2 = 8$.

Level 2

Why is 2 cubed different from 2 times 3? Evaluate both and explain what makes the answers different.

- 2 cubed equals ____ but 2 times 3 equals ____ because ____.
- An exponent means ____, while multiplication means ____.

CONNECT

When you bounce a track, the file size grows as you double the layers. Where else do you see quantities that grow by repeated multiplication?

Level 1 support

Start here: Doubling money, folding paper, or sharing a rumor all grow by repeated multiplication.

 **Need a hint?**

Sentence starters (optional)

I see repeated multiplication when ____.

Veo multiplicación repetida cuando ____.

That situation could be written with a ____ because ____.

Esa situación se podría escribir con una ____ porque ____.

Word bank: base exponent power evaluate growth

Listen for: Students give a real doubling or repeated-multiplication example and connect it to a base and exponent.

Level 2

Compare 3 to the 4th power and 4 to the 3rd power. Are they equal? Evaluate both and use base and exponent to justify your answer.

- 3 to the 4th equals ____ and 4 to the 3rd equals ____, so they are ____.
- Switching the base and exponent changes the value because ____.

Write About the Math

The Writing Revolution

I can explain my work using the words exponent, base, power, and evaluate.

1. Kernel Sentence subject + verb

Model: Exponent is a small number that tells how many times to multiply the number by itself.

Exponente es un número pequeño que dice cuántas veces multiplicar el número por sí mismo.

Write a kernel sentence about exponent. Use a subject and a verb.

Escribe una oración base sobre exponente. Usa un sujeto y un verbo.

2. Sentence Expansion because · but · so

Kernel: Exponent matters in math

Exponente importa en matemáticas

Expand the kernel three ways. Add a reason, a contrast, and a result.

because
porque

Exponent matters in math because ____.

Exponente importa en matemáticas porque ____.

but
pero

Exponent matters in math, but ____.

Exponente importa en matemáticas, pero ____.

so
entonces

Exponent matters in math, so ____.

Exponente importa en matemáticas, entonces ____.

3. Sentence Types 4 ways to write a math idea

Statement
Afirmación

Tell one true fact about exponent.
Di un hecho verdadero sobre exponent.

Exponent ____.

Question
Pregunta

Ask a question about exponent.
Haz una pregunta sobre exponent.

How does ____ ?

¿Cómo ____ ?

Exclamation
Exclamación

Show excitement about exponent.
Muestra entusiasmo sobre exponent.

Wow, ____ !

¡Guau, ____ !

Command
Mandato

Tell a partner what to do with exponent.
Dile a un compañero qué hacer con exponent.

First, ____ .

Primero, ____ .

4. Explain Your Reasoning use a sentence starter

Each number is the one before it times ____ , **so it is repeated** ____ .

Cada número es el anterior por ____ , *así que es* ____ *repetida.*

2 times 2 times 2 can be written as 2 to the ____ **power.**

2 por 2 por 2 se puede escribir como 2 a la ____ *potencia.*

The base is ____ **and it is the number being** ____ .

La base es ____ *y es el número que se está* ____ .

Try It

Solve on your own. Check the answer key when you are done.

1. Evaluate: $3 \times (4 + 2)^2$

- A. 108
- B. 54
- C. 36
- D. 144

Show your work:

Stretch Your Thinking

Level 2 enrichment

Challenge task — explain your reasoning in full sentences.

A bacteria colony doubles every hour. Starting with 1 bacterium, write a power expression for the number after 6 hours. Explain why exponential growth is so much faster than adding the same number each hour.

Sentence starter: After 6 hours there are ____ bacteria because $2^6 =$ _____. If it grew by adding 2 each hour instead, there would only be ____ bacteria. Exponential growth is faster because _____.

Show your work:

Reflect — Exit Ticket

What is the value of 3^4 ?

A. 81

B. 12

C. 34

D. 64

Your answer:

Answer Key & Teacher Guide

1. **We Try (notes):** A. $36 - 6^2 = 6 \times 6 = 36$.
2. **You Try (notes):** A. $5 \times 5 \times 5 - 5^3$ means use 5 as a factor 3 times: $5 \times 5 \times 5$. The exponent tells how many times to multiply, not what to multiply by.
3. **Try It 1:** A. $108 - 4 + 2 = 6$. $6^2 = 36$. $3 \times 36 = 108$.
4. **Exit Ticket:** A. $81 - 3^4 = 3 \times 3 \times 3 \times 3 = 81$.

Writing (TWR) — what to look for

- **Kernel sentence:** A complete sentence needs a subject and a verb. Example: Exponent is a small number that tells how many times to multiply the number by itself.
- **Expansion:** *because* gives a reason, *but* shows a contrast or exception, *so* shows a result. Answers vary; each must keep the kernel idea and add the correct kind of detail.
- **Sentence types:** Statement ends with a period, question with "?", exclamation with "!", and a command starts with an action verb (a "bossy" verb).